Medha Srivastava

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Dept. of Mechanical Engineering

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ACADEMIC QUALIFICATIONS

Year	Degree/Certificate	Institute	CPI/%
2025 (expected)	B.Tech	Indian Institute of Technology Kanpur	7.52/10
2020	CBSE(XII)	R.S.J Public School	94.2%
2018	ICSE(X)	St. Mary's Convent Inter College	93.6%

AWARDS & ACHIEVEMENTS

- Selected for **Stage 2** of India's biggest Robotics Competition, **ROBOFEST-GUJARAT 3.0**, showcasing exceptional ideation skills
- Amongst the **top 200** out of **3000+** applicants across India to be selected for the SURGE 2023 research internship program
- Achieved **All India Rank 3189 (Top 0.3%)** in Joint Entrance Exam (JEE) Main 2021 out of over **1,000,000 +** candidates
- Achieved **All India Rank 6023** in Joint Entrance Exam (JEE) Advanced 2021 out of a pool of **260,000+** selected candidates

WORK EXPERIENCE

Research Intern | SURGE'23 | **Mentor:** Prof. Shikhar Krishn Jha *(May'23 - July'23)*

Objective	• **FEM** Analysis and modelling of Flash Sintering phenomenon in 8YSZ Material using **COMSOL Multiphysics**
Approach	• Implemented a comprehensive approach by utilizing FEM to accurately simulate the **flash sintering** process in **8YSZ** material, considering relevant equations, initial and boundary conditions, leading to insightful analysis
Impact	• Achieved superior temperature distribution model of **20-second** simulation as compared to the original 3-second • Obtained a linearized stress-strain graph for **98 %** dense **8YSZ** sample by applying a boundary stress of **300 MPa** • Developed a model demonstrating a **42% increase** in core temperature **(1123 K to 1595 K)** during Joule heating

Business Analyst Experience Program | Finlatics *(May'23 - Jun'23)*

Objective	Develop expertise in business research and conduct data analysis with **Excel** and **Power BI** to enhance proficiency.
Approach	Created issue tree using **MECE analysis**, evaluated acquisitions for niche technologies and assessed growth factors
Impact	Conducted strategic analysis on **mergers and acquisitions** and achieved an increase of **12%** in y-o-y profit margin

KEY PROJECTS

Project Mentor : Forecasting Using Time Series Analysis | Stamatics IIT Kanpur *(May'23 - July'23)*

- Mentored **100+** shortlisted students in a **8-week** long project focussed on **Time Series Analysis** using the **Statsmodel** library
- Applied ADF test, ACF test, and **Box-Cox transformation** for stationarity, normality and homoscedasticity on **7+ time series**
- Analyzed air pollution with visualizations, and constructed an accurate **order 10** moving average model with **0.998 R2 score**

ML Appication in Mechanical Engineering | Association of Mechanical Engineers (AME) IIT Kanpur *(Dec'22 - March'23)*

- Utilized **vibration sensors** data of varying unbalance strengths and **rotation speeds** to detect unbalances in rotating machinery
- Implementing various machine learning algorithms like linear, logistic regression, **SVMs, decision trees, random forests, KNN**
- Developed an ML model for **unbalance detection** in rotating shafts using vibration data using **FFT preprocessing** techniques.
- Evaluated the performance of **FCN** with varying numbers of hidden layers achieving a maximum accuracy of **98.6 %**

Image Processing and Computer Vision | Stamatics IIT Kanpur *(May'23 - July'23)*

- Acquired knowledge of probability distributions, **Image sampling**, quantization, enhancement, and edge detection techniques
- Applied spatial domain transformation techniques and filtering methods like **adaptive threshold** and **histogram equalization**
- Implemented Module design and achieved **79% accuracy** in multi-class image classification using **deep learning** model

Clutchnaut | Robotics Club IIT Kanpur *(Oct'22)*

- Built a transmitter-controlled gripping bot for **Techfest** designed to trace arena according to the problem statement **Cozmoclench**
- Utilized SERVO and DC motors, l298 motor driver,**Arduino Mega** as the microcontroller and **transmitter-receiver** for controls

Physilearn | Stamatics IIT Kanpur *(May'23 - July'23)*

- Learned how **Physics Informed Neural Networks** combine NN and differential equations to predict system behavior
- Implemented a PINN architecture utilizing popular Python libraries like Matplotlib, Pandas, NumPy, and sci-kit learn
- Acquired practical skills by applying **PINNs** in PyTorch for tasks like **harmonic oscillator** modeling and heat equation solving

TECHNICAL SKILLS

Languages : SQL, C, C++, Python **Libraries & Softwares** Numpy ,pandas , OpenCV, COMSOL, MATLAB, Arduino , AutoCAD

RELEVANT COURSES

(Online Ongoing^o)*

Fundamentals of Computing	Probability and Statistics*	Machine Learning*	Image Processing^o
Rigid Body Dynamics	Fluid Mechanics	Thermodynamics	Mechanics of Solids
Nature Properties of Materials	Engineering Graphics	Introduction to Electronics	Data Structures & Algorithms

POSITIONS OF RESPONSIBILITY

Manager, UG Academics Wing | Academics and Career Council, IIT Kanpur *(May'23 - Present)*

- Recruited a team of **12 secretaries**, **62 Department Mentors** and **28 student nominees** to address student departmental issues.
- Conducted **Academic Orientation** for **1200+** UG freshers, acquainting them with the academic curriculum & **UGARC** reforms
- Managed budget of **Rs.1.5 Lakhs+** for sessions, booklets and optimized expenses, achieving **10% cost savings**
- Ensured proper communication between **5000+** students and authorities for pertinent academic doubts & curriculum changes

Secretary | Robotics Club | IIT Kanpur (*July'22-May'23*)

- Represented IIT Kanpur and robotics club at **Techfest 2022** (IIT Bombay) by participating in the **Cozmoclench** competition
- Organized **SNT Pavilion** for Y22 freshers, educating **1000+** students about various bots and their technical aspects.

EXTRA-CURRICULAR ACTIVITIES

- **Winner** of the **Most Suave Presentation Award** in **Case-O-Mania 2021** by Finance and Analytics Club, IIT Kanpur.
- Conducted the **MATHEMANIA 2022** (flagship event of Stamatics) with more than 500 participants from all over the country.